Ajay Venkatesh

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Skills

Languages: TypeScript, JavaScript (ES6), HTML5, CSS3, SQL
Frameworks & Libraries: Angular, RxJS, NgRx, React, Redux, Node.js, Express.js
Testing & Performance: Jest, Jasmine, Karma, React Testing Library – Lazy Loading, Code Splitting, Client-Side Caching, Responsive Design
APIs & Tools: REST APIs, GraphQL, WebSockets, Git, AWS (Lambda, S3, EC2, API Gateway, Cognito), Azure, Docker, CI/CD
AI/ML: LLM Fine-tuning, RAG (LangChain, FAISS), NLP, Deep Learning.

Experience

Campus Mesh — Jan 2026 – Present

Software Engineer | *Angular, RxJS, real-time microservices, and LLM-based AI retrieval pipelines*

- Engineered reactive **Angular** components and services using **RxJS** to handle real-time academic workflows – study groups, messaging, and notifications – serving **10,000+ active users** with consistent uptime.
- Cut the initial JavaScript payload by **35%** by deferring feature module downloads to on-demand route navigation, eliminating upfront loading of pages the user may never visit and improving Time to Interactive on slow connections.
- Raised unit and integration test coverage to **85%+** using **Jest** and **Jasmine/Karma** across all feature modules, enabling weekly deployments with zero production regressions through **Git**-managed CI pipelines.

Amazon — May 2025 – Aug 2025

Software Development Engineer Intern | *React 18, TypeScript, serverless AWS infrastructure, and Java backend systems*

- Architected a **React 18** and **TypeScript** console with **OAuth 2.0** and corporate **SSO** authentication; enforced **4-tier role-based access control** governing which routes, UI panels, and actions each user could access — protecting evidence data for **500+** fulfillment team members.
- Eliminated redundant API calls by caching server responses on the client for high-traffic dashboard views; applied page-level code splitting so each route loaded only its required bundles — cutting page load speed by **40%** and improving Core Web Vitals scores.
- Maintained zero critical defects across a 3-month release cycle by writing unit and integration tests covering auth flows, role guard behavior, and data-fetching components using **Jest** and **React Testing Library**.

New York University (Novel AI Technologies, Inc.) — Aug 2024 – Present

Software Developer (On-Campus) | *React, TypeScript, REST APIs, WebSockets, and cloud-deployed AI health platform*

- Built a **TypeScript** and **React** frontend for an **NSF-funded ($100K)** AI health platform, integrating real-time wearable sensor data with cloud-deployed machine learning models for iOS and Android users.
- Delivered an insurance analytics platform (**React, Node.js, REST APIs**) where brokers, underwriters, and admins each accessed role-restricted dashboards scoped to their permitted data; integrated **WebSocket** real-time updates, **responsive HTML5/CSS3** layouts, and Stripe billing — generating **$50K+** across **6 enterprise clients**.
- Built a shared **React** component library with consistent design standards, cutting UI development time across 3 product features; coordinated **Git** branching and code review workflows across a multi-developer team.

PricewaterhouseCoopers (PwC) — Aug 2021 – Aug 2024

Senior Software Engineer | *PCF controls, Angular/TypeScript UI, HTML/CSS, and Azure cloud-native backend*

- Designed and delivered custom **Power Apps Component Framework (PCF)** controls in **TypeScript, HTML5, and CSS3** for **Dynamics 365 CRM**, replacing out-of-box UI with accessible data interaction components across **3 enterprise** deployments.
- Built **Angular** and **TypeScript** frontends integrated with Microsoft CRM, handling asynchronous, event-driven data streams across health platform modules with reactive state management.
- Established frontend code review standards covering **HTML/CSS** accessibility, component design patterns, and test coverage; reduced production defects by **25%** and mentored **3 junior engineers**.
- Deployed an **NLP-based OCR** model on Azure, cutting data extraction time by **75%**; built **Power BI** dashboards across multiple databases, improving reporting efficiency by **40%**.

Projects

Task Management Dashboard (Angular SPA) | ***Angular, TypeScript, RxJS, NgRx, HTML5/CSS3, Jest***

- Built a single-page **Angular** application with **NgRx** for centralized state management and reactive data streams; delivered drag-and-drop task boards and real-time status updates via WebSockets.
- Achieved **90%+ test coverage** with **Jest** across components, services, and state management layers; enforced strict **TypeScript** typing and type-safe selectors throughout.

Hospital Management System (Full-Stack Web Application) | ***React, TypeScript, Node.js, Express, MongoDB, AWS SageMaker***

- Developed a full-stack healthcare platform with **React** and **TypeScript**, featuring responsive **HTML5/CSS3** layouts and role-based access control to manage appointments, patient records, and lab reports.
- Fine-tuned a **Hugging Face** transformer model to **88% diagnostic accuracy** and deployed it on **AWS SageMaker**, enabling real-time condition predictions within the physician dashboard.

Education

New York University | *Master of Science in Computer Engineering* ***(GPA – 3.9/4)*** — Sep 2024 – May 2026

- *Coursework:* System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures

Certifications

AWS Cloud Practitioner
Azure Data Fundamentals
Microsoft Solution Architect Expert
Power BI Data Analyst Associate
Microsoft Developer Associate
Stanford Machine Learning (Coursera)